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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B.Tech (CBCS) Degree Examinations

December 2025

CC2103-1 – FUNDAMENTALS OF COMPUTER ARCHITECTURE AND ORGANIZATION

(For CCE)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. In signed 4-bit addition, what happens when $0110 + 0111$ is performed?
A) Result = 1101, Overflow occurs B) Result = 10101, No overflow
C) Result = 1101, No overflow D) Result = 11101, Overflow occurs
2. Multiply $(-5) \times (+6)$ using Booth's algorithm (4-bit). The result is:
A) 00010010_2 B) 00010000_2
C) 00011110_2 D) 00011100_2
3. The time required to complete one instruction is
A) Fetch time B) Execution time
C) Control time D) Complete time
4. Which method is used to detect double errors and pinpoint erroneous bits
A) Even Parity check method B) Odd parity
C) Checksum D) Error-correcting code
5. The partial remainder is restored by adding the divisor to the negative difference
A) In non-restoration Division B) In restoration method
C) In comparison method D) In decimal arithmetic method
6. _____ holds the address of the instruction to be executed next and is incremented each time an instruction is fetched from memory
A) MAR B) PC
C) MDR D) IR
7. Which of the following best describes Von Neumann architecture?
A) Separate memory for instructions and data B) Shared memory for instructions and data
C) Instructions only in CPU registers D) Instructions hardwired into control unit
8. In hardwired control, control signals are generated by
A) Microprograms B) Combinational circuits
C) Firmware D) Software
9. In immediate addressing mode, the operand is
A) In a register B) In memory
C) Inside the instruction itself D) At address pointed by another address
10. Which micro-operation performs logical AND between registers R1 and R2?
A) $R3 \leftarrow R1 + R2$ B) $R3 \leftarrow R1 - R2$
C) $R3 \leftarrow R1 \wedge R2$ D) $R3 \leftarrow R1 \vee R2$

11. Microprogrammed control unit generates signals using
 A) Control memory B) Hardwired circuits
 C) Instruction register D) Decoder only
12. Interrupts are used to
 A) Speed up execution B) Handle external events
 C) Replace memory fetch D) Decode instructions faster
13. The advantage of using three-address instructions is
 A) Smaller program size B) Simpler decoding
 C) Faster memory fetch D) Reduced opcode bits
14. Indexed addressing is commonly used for
 A) Jump instructions B) Array access
 C) Subroutine calls D) Interrupt handling
15. Cache memory improves performance by
 A) Storing secondary data B) Using RAID
 C) Exploiting locality of reference D) Removing interrupts
16. Page fault occurs when
 A) Page is in cache B) Page not in memory
 C) RAM is full D) CPU halts
17. Thrashing occurs when
 A) Too many interrupts B) Excessive paging
 C) DMA fails D) Cache is empty
18. RAID level that uses mirroring is
 A) RAID 0 B) RAID 1
 C) RAID 5 D) RAID 6
19. Programmed I/O requires
 A) CPU intervention for every transfer B) DMA
 C) Interrupts D) No CPU intervention
20. Polling is
 A) Device-driven B) The CPU continuously checks the device status
 C) Always faster than interrupts D) Used in RAID

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) With a neat diagram, discuss the operation concepts in a computer highlighting the role of PC, MAR, MDR & IR.	8	L*2	1	1,3,9,12
	b) Given the multiplicand A = +8 and multiplier B = -10. Solve the multiplication of A and B using Booth's algorithm.	8	L3	2	3,9,12
2.	a) Explain with an example, the flowchart for addition and subtraction in signed-2's complement representation.	8	L2	1	1,3,9,12
	b) Perform the addition and subtraction on the following pairs of numbers using 5-bit signed 2's complement format. Determine about overflow in each case i) +12 and +9 ii) -15 and -9 iii) +10 and -8 iv) -2 and +7.	8	L3	1	1,3,9,12
3.	a) Solve, $-14 \div +4$ using restoring division method. Elaborate each step.	8	L3	2	3,9,12
	b) Explain error detection codes in detail with examples.	8	L2	2	3,9,12
Unit – II					
4.	a) Analyze the execution of BUN instruction using flowchart and micro-operations.	8	L2	3	3,9,12
	b) Demonstrate the direct and indirect addressing with a neat diagram and an example.	8	L2	3	3,9,12

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| 5. | a) | Illustrate the flowchart of the instruction cycle, including the fetch, decode, and execute phases. | 8 | L2 | 3 | 3,9,12 |
| | b) | Discuss the input, output and interrupt with an example. | 8 | L2 | 4 | 9,12 |
| 6. | a) | What is interrupt? Explain the flowchart for interrupt cycle. | 8 | L2 | 4 | 9,12 |
| | c) | Write a program to evaluate the arithmetic statement: | | | | |

$$X = \frac{A - B + C * (D * E - F)}{G + H * K}$$

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|----|--|---|----|---|------|--|
| a. | Using a general register computer with three address instructions. | | | | | |
| b. | Using a general register computer with two address instructions. | 8 | L3 | 4 | 9,12 | |

Unit – III

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|----|----|--|---|----|---|------|
| 7. | b) | Discuss demand paging and page replacement policies. | 8 | L2 | 5 | 9,12 |
| | c) | Explain the cache coherence problem in multiprocessor systems. How does it affect system performance and consistency? | 8 | L2 | 5 | 9,12 |
| 8. | a) | Analyze a situation where programmed I/O is causing a bottleneck in a system. Propose a solution using DMA to improve performance. | 8 | L2 | 5 | 9,12 |
| | b) | Explain the concept of RAID (Redundant Array of Independent Disks). Discuss the different RAID levels and their trade-offs in terms of performance, data redundancy, and cost. | 8 | L2 | 5 | 9,12 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
